A family of quartics in 317 variables from aggregate trace heights

8 November 2026

Abstract

We specify a family of integer quartics in 317 total variables by 291 quadratic equations and a complete coefficient table polynomial in one fixed integer parameter. Under the arithmetic premises of the specified source construction, every sufficiently large member defines the integers in the rationals. The new ingredient is a linear height comparison for the sum of four rational traces of elliptic abscissas. It reduces three height-detector blocks to two and replaces a field product by a rational sum. The absence of a parity restriction also removes all four odd-index certificates. An input relabeling retains exact integer extraction. The sufficient parameter threshold is ineffective because the height comparison has an ineffective additive constant. Thus this is an existence theorem with an exact family specification; it does not determine the coefficients of a particular defining member. The fully specified 389-variable quartic remains a separate result.

1 Statement and exact family

Every variable count includes the input X. Put $\ell = 80989$ and

$$A_N = \ell^{2N+1} \qquad (N \geq 1).$$

The files `quadratic-family317.json` and `polynomial-family317.json` specify 291 integer quadratic polynomials $Q_{N,j}$ in 317 coordinates and

$$P_N = \sum_{j=1}^{291} Q_{N,j}^2.$$

The table uses powers of A_N as a compact notation for integer coefficients. Neither A_N nor N is a variable of P_N.

Theorem 1.1. *Assume the arithmetic input contract of the specified quartic source [1, Section 2], including its quantitative weighted prime-selection premise. There is an integer $N_0 \geq 27464$ such that, for every integer $N \geq N_0$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{316},\ P_N(x, \boldsymbol{y}) = 0.$$

For every $N \geq 1$, the polynomial has total degree exactly four, integer content one, and coefficient height

$$H(P_N) < 2^{1520}\, 80989^{4N+2}.$$

For $N \geq 27464$ it has exactly 5032 nonzero coordinate monomials. The proof does not determine a numerical value of N_0.

The height comparison used below is a proved consequence of Von Buhren's theorem, not a new arithmetic conjecture. Its ineffectiveness prevents us from selecting an explicit defining member from the family. Finite coefficient verification establishes the specified polynomial identities; it cannot resolve this effectiveness issue or prove the retained arithmetic premises.

2 The aggregate trace

We use the same curve and point as the source:

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d = 3 - \rho, \qquad E : y^2 = x^3 - d^2 x,$$

$$P = 2(1 + \rho, 2\rho) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32} \right).$$

The retained root charts give four points

$$Q_1 = [n]P, \quad Q_2 = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P$$

with nonzero integer labels, as proved below. Define the rational number

$$F = \frac{1}{2} \operatorname{Tr}_{K/\mathbb{Q}}(x(Q_1) + x(Q_2) + x(Q_3) + x(Q_4)), \qquad B = \max(|n|, |a|, |t|, |b|).$$

We write h for absolute logarithmic Weil height and $\widehat{h}_E$ for canonical height on E.

Lemma 2.1. *Assume $2E(K) = \mathbb{Z}P$ and $P = 2R_0$, where $R_0 = (1 + \rho, 2\rho)$. Every solution in K of*

$$x = h^2, \qquad j^2 = x^2 - d^2$$

gives a point $Q = (x, hj) = [m]P$ for a nonzero integer m. No parity condition on m is imposed.

Proof. If $h = 0$, then $j^2 = -d^2$, impossible in the real field K. Thus $x \neq 0$, and the equations imply that Q lies on E. The standard squareclass homomorphism for this two-isogeny is

$$\delta_x : E(K) \longrightarrow K^\times / K^{\times 2}, \qquad \delta_x(Q) = [x(Q)]$$

away from O and $T_0 = (0, 0)$, with $\delta_x(O) = 1$ and $\delta_x(T_0) = [-d^2]$. For completeness, its homomorphism identity follows from the chord-and-tangent rule: the product of the three intersection abscissas of a nonvertical line with the curve is the square of its ordinate intercept. The special value at T_0 agrees with $x(Q + T_0) = -d^2/x(Q)$; the vertical and tangent cases give the same identity.

The group premise gives $2Q = mP = 2mR_0$, and hence $Q = mR_0 + T$ for some $T \in E[2](K)$. The four torsion points are $O, T_0, (d, 0), (-d, 0)$, with squareclasses $1, -d^2, d, -d$. They all have uniform real signs at the two embeddings. In contrast, $\delta_x(R_0) = [1 + \rho]$ has mixed real signs. Since $x(Q)$ is a nonzero square, m must be even. Then $\delta_x(T)$ must be a square. The negative values $-d^2, -d$ are not squares, and d is not a square because its norm to $\mathbb{Q}$ is 7. Therefore $T = O$ and $Q = [m/2]P$. The index is nonzero because the point supplied by the chart is finite. This also proves it is nontorsion. □

Lemma 2.2. *There are fixed constants $M_1 > 0$ and $M_2 \geq 1$ such that*

$$B^2 \leq M_1 h(F) + M_2.$$

Also $F > 12$, so 1 and F^2 are distinct positive rational squares. No effective value of M_2 is claimed here.

Proof. The aggregate-trace height theorem [3] gives

$$\sum_{i=1}^{4} \widehat{h}_E(Q_i) \leq M h(F) + M_0$$

for fixed constants. The point P has positive canonical height and $\widehat{h}_E([j]P) = j^2 \widehat{h}_E(P)$; rescaling proves the first claim. The theorem applies to all nonzero integer labels and requires no parity

condition. Its proof excludes every positive-dimensional abelian translate through the indicated points in a fiber of the aggregate map. Von Buhren's family height theorem then supplies the linear inequality, with an ineffective additive constant.

Each Q_i is a nonzero double. Its abscissa is a positive square at both real embeddings; the curve equation gives $x(Q_i) > d$ at each embedding. Since d and its conjugate sum to 6, the rational trace of each abscissa is greater than 3. Summing gives $F > 12$. □

3 Two height detectors

Use only the scales $1, F^2$. Every retained local detector has the same point tests, formal logarithm approximations, and test orders as the 389-variable construction [2]. Its three classes of test orders are $1, 10, 40$. In particular, its nonvanishing is automatic at every consumed contact prime by the local argument of that construction; no trace inverse witness is reintroduced.

Let H denote the retained contact-height constant and let M_0, M_1^* be the two contact radicals. We distinguish the radical notation M_1^* from the constant M_1 in Lemma 2.2. The following implication is the only height consequence required of the detectors.

Lemma 3.1. *If $J \neq 0$ is an integer and both M_0^k and $(M_1^*)^k$ divide J, then*

$$h(F) \leq (H + \log 3)|J|^{19/k}, \qquad B^2 \leq C|J|^{19/k} + M_2, \quad C = M_1(H + \log 3).$$

Proof. The retained contact-height theorem bounds both $h(\lambda)$ and $h(F^2\lambda)$ by $(H + \log 3)|J|^{19/k}$. The factor $\log 3$ accounts for the fractional linear change in the contact argument. Since the detector domain excludes $\lambda = 0$,

$$2h(F) = h(F^2) \leq h(F^2\lambda) + h(\lambda).$$

This proves the first inequality. Apply Lemma 2.2 for the second. □

4 Soundness for every sufficiently large parameter

Put

$$\beta = 51X^2 + 2504, \qquad c = |a - 1|, \qquad \Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1.$$

The scalar block uses

$$r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1), \qquad R = \frac{A_N r}{1 + \ell^2 r}, \qquad c_R = R + 9792.$$

All factors of r are positive for rational X. The constant 9792 replaces the old shift 4896; its prime factors are still only $2, 3, 17$. The five excluded squareclasses at ℓ are unchanged, because the old affine input $2X_{\text{old}} + 1$ is now X. Thus the same Euler tests give

$$v_\ell(R) \in \{2N + 1, 2N - 1\}.$$

At every prime in $M_5(\beta)$, R still has positive odd valuation and c_R is a unit. Consequently the local scalar argument, including the exceptional-point exclusions, is unchanged. Its integral group identity is $[m](2G) = [4(a-1)]H$ and requires only $a - 1 \neq 0$ integral, not even. It gives, for every $N \geq 1$,

$$c \geq \ell^{L_N}, \qquad L_N = \left\lfloor \frac{187(2N-1)}{8} \right\rfloor + 1, \qquad h(\beta) \leq Hc^{19/24}. \tag{1}$$

These conclusions are proved before the final height comparisons; they do not depend on the number of height detectors. In particular c tends uniformly to infinity with N, and $B \geq |a| \geq c - 1$. All exceptional elliptic-point exclusions and ratio-domain checks from the scalar argument remain in force.

Lemma 4.1. *For every sufficiently large N, every rational solution of the quadratic system satisfies $\Delta = 0$.*

Proof. If $\Delta \neq 0$, the retained local comparisons make the fortieth powers of both contact radicals divide Δ. For $B \geq 1$, the displayed expression gives

$$|\Delta| \leq 2B^4 + 6B^3 + 5B^2 + 1 \leq 14B^4.$$

Lemma 3.1 would therefore imply

$$B^2 \leq C\, 14^{19/40} B^{19/10} + M_2.$$

Dividing by B^2 shows this is impossible once B exceeds a fixed threshold, since $19/10 < 2$. Equation (1) ensures that threshold for every sufficiently large N. □

Lemma 4.2. *For every sufficiently large N, every rational solution satisfies $n = \beta$ and $X \in \mathbb{Z}$.*

Proof. The elementary Lucas growth lemma in [1, Section 5] states that $\Delta = 0$, $b \neq 0$, and $|a| > 7$ imply

$$\log B \geq \log |t| > |a| - 3.$$

Suppose $n - \beta \neq 0$ and let J be its nonzero reduced numerator. The order-ten tests make both tenth powers of the contact radicals divide J. Also $|J| \leq 2B \exp h(\beta)$. By Lemma 3.1,

$$B^2 \leq C\, 2^{19/10} B^{19/10} \exp\!\left(\frac{19}{10} h(\beta)\right) + M_2.$$

For $B^2 \geq 2M_2$ it follows that

$$\log B \leq 10 \log(2C) + 19 \log 2 + 19 h(\beta) \leq 10 \log(2C) + 19 \log 2 + 19 H c^{19/24}.$$

The last expression is $o(c)$ as $c \to \infty$, whereas $|a| - 3 \geq c - 4$. This contradicts the Lucas lower bound once c exceeds a fixed threshold. Equation (1) provides that threshold for every sufficiently large N.

Thus $\beta = n$ is an integer. If $X = u/v$ in lowest terms with $v > 0$, then $v^2 \mid 51$. Since 51 is squarefree, $v = 1$, so $X \in \mathbb{Z}$. No parity restriction on n is used. □

The finitely many thresholds in these arguments depend on the fixed constants C, M_2, H. Since L_N increases with N, there is one N_0 such that all conclusions hold for every $N \geq N_0$. Increase it to be at least 27464 if necessary. This proves the soundness implication with the quantifiers asserted in Theorem 1.1. The dependence on M_2 is precisely the ineffective part.

5 Completeness

Fix $N \geq 1$ and an integer input X. The unchanged scalar normalizer and point construction admit the same prescribed finite local depths and real approximation conditions as the source. Increasing N changes a fixed coefficient and finite depth requirements; it does not add a rational variable or an arithmetic existence premise. The source proof permits these requirements to be fixed before the later contact primes are chosen.

Here are the required small-prime conditions for the modified label and shift. Directly from $\beta = 51X^2 + 2504$ one obtains

$$v_2(r) = \begin{cases} 3, & 4 \mid X, \\ 2, & 4 \nmid X, \end{cases} \qquad v_3(r) = 1, \qquad r \equiv 15 \pmod{17}.$$

For the first assertion, odd X gives $\beta \equiv 3 \pmod 8$, with two valuation-one factors in r. If X is even, the only even factor is $\beta = 4(51(X/2)^2 + 626)$, of valuation two or three as displayed. At three the only divisible factor is $2\beta - 1 = 3(34X^2 + 1669)$, whose quotient is nonzero modulo three. At seventeen, $\beta \equiv 5$ and direct substitution gives the stated residue.

The denominator $1 + \ell^2 r$ is a unit at two and three and is 16 modulo seventeen; A_N is a unit at all three primes. Since $9792 = 2^6 \cdot 3^2 \cdot 17$, this gives

$$v_2(9792/R) \geq 3, \quad v_3(9792/R) = 1, \quad v_{17}(9792/R) = 1.$$

Hence $c_R/R = 1 + 9792/R$ is a square in each of $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. The scalar pole input $c_R R W^2$ is therefore locally square, and the same simultaneous depth and real approximation argument makes it lie in $(0, 17)$ with a square reduced denominator and all required local depths.

The input root index $n = \beta$ may now be even. The comparison construction permits every nonzero integer label: its points are $U_i = [hn_i]P$ and $V_i = [(h-1)n_i]P$, and the simultaneous congruences for h are independent of the n_i. The scalar and Lucas choices for the other labels can retain their former parities. Every such root has the square-abscissa chart of Lemma 2.1; the four additional odd-index charts are no longer required.

After the four root points have been chosen, F is a fixed positive rational number. The two scales $1, F^2$ are distinct positive rational squares by Lemma 2.2. The retained simultaneous prime-pattern and pole realization argument applies to this fixed finite list. Coincident zero contact offsets are the common pattern already present in the original construction. The nonzero contact offsets from the two scales do not collide: the ratio of any two nonzero elements of

$$\left\{-\tfrac{1}{2}, -1, \tfrac{1}{2399}, -\tfrac{32}{407}\right\}$$

with the same sign has absolute value at most $407/32 < 144 < F^2$. All required avoidance congruences are incorporated in the same finite-support selection. The quantitative separation argument retains its stated weighted premise.

The resulting witnesses satisfy all retained comparisons, the two detectors, and their normalized poles. Their quadratic gate values are obtained by the same rational formulas; the local inverse guards are omitted as in [2]. The new scale coordinate is $F^2/4$. This supplies a rational solution for every integer input. Finally, a sum of rational squares vanishes exactly when all its summands vanish. Together with the soundness argument, this proves the projection assertion in Theorem 1.1.

6 Exact coordinate accounting and verification

Start with the specified 389-variable system, whose full source hash is recorded in the generator. Delete the second component height detector, including its pole certificate and its private evaluation coordinates. This removes 52 coordinates and 50 quadratic rows. Of the nine field-product and scale coordinates, retain only the coordinate formerly called `heightProduct.scale0.quadratic`; the other eight and their defining rows are deleted. Rename the retained coordinate s_F and replace its one row by

$$(x^*_{A,0} + x^*_{S,0} + 2x^*_{T,0} + x^*_{Z,0})^2 = 4s_F.$$

The starred symbols are literal stored coordinates. Their combination on the left before squaring is exactly F; in particular the factor two on the third coordinate is required by its existing normalization. The retained first component detector uses the actual input $4\lambda s_F = F^2\lambda$. Every one of its downstream rows is unchanged. The full affine detector normalization is independently verified in [2].

Delete the three private odd-index coordinates and their three rows for each of the four roots. There are no other occurrences of these twelve coordinates. The retained square-abscissa charts are sufficient by Lemma 2.1.

Finally substitute into every retained row

$$X_{\mathrm{old}} = \frac{X-1}{2}, \qquad Y_{\mathrm{old}} = \frac{Y-2X+1}{4},$$

where Y_{old} was the stored square of the old input. The old square row becomes $Y = X^2$, and

$$51(2X_{\mathrm{old}}+1)^2 + 2504 = 51X^2 + 2504.$$

The transformation is affine and invertible on the full coordinate space: $X = 2X_{\mathrm{old}} + 1$ and $Y = 4Y_{\mathrm{old}} + 4X_{\mathrm{old}} + 1$. Thus it preserves quadratic degree. Replace the scalar shift 4896 by 9792 in its one defining row. Clear each row's rational denominators and divide its integer coefficient content; neither operation changes its rational zero set.

The counts are therefore

$$389 - 52 - 8 - 12 = 317, \qquad 361 - 50 - 8 - 12 = 291.$$

The independent checker applies the inverse affine map to every new row and recovers the corresponding old row, including the changed scalar constant, up to a specified nonzero rational multiple. It separately checks sole uses of every deleted coordinate and the complete sum-of-squares coefficient table. A parameter-independent coefficient one certifies content one and degree exactly four. The small-prime congruences and the five Euler tests at 80989 are also recomputed independently.

There are 5047 terms when powers of A_N are listed separately, and 5032 coordinate monomials after collection. Every coefficient is an integer polynomial in A_N of degree at most two. The largest sum of absolute coefficient-polynomial entries is less than 2^{1520}, giving the height bound in the theorem. For $A_N > 2^{1521}$, every nonconstant coefficient polynomial has its leading term larger than the sum of the lower terms. Thus none vanishes there; $N \geq 27464$ more than suffices.

The complete reproduction sequence, in the construction directory, is

```
python3 build_quartic389.py
python3 check_quartic389.py
python3 verify_detector_normalizations.py
python3 build_quartic329.py
python3 check_quartic329.py
python3 build_quartic317.py
python3 check_quartic317.py
```

The intermediate family is only a specified algebraic stage in the construction. The exact final family and verification reports distinguish the algebraic specification from the ineffective existence threshold.

References

[1] J. Brox and contributors, *A forced prime condition and an explicit quartic in 392 variables*, 8 November 2026, commit `de7aa60dd1794d7ea62a8a83cdc4bfd22380c74f`, https://github.com/JoseBrox/diophantine-z-q_improvements.

[2] J. Brox and contributors, *A quartic in 389 variables from local nonvanishing*, 8 November 2026, accompanying proof and exact coefficient data.

[3] J. Brox and contributors, *Aggregate trace heights of elliptic multiples*, 8 November 2026, accompanying proof.